

Exploratory Data Analysis (EDA) – Practical Tasks

Dataset: Employee Sales Data

The following dataset contains employee and sales-related information. Use this data to perform EDA using NumPy, Pandas, and Matplotlib.

Emp_ID	Age	Gender	Department	Experience	Salary	Sales
101	25	F	HR	2	22000	150000
102	30	M	Sales	5	35000	320000
103	28	F	IT	4	40000	280000
104	35	M	Sales	8	55000	450000
105	40	M	HR	10	60000	200000
106	29	F	IT	3	38000	300000
107	32	M	Sales	6	48000	410000
108	27	F	HR	2	21000	160000

EDA Tasks

1. Load the dataset into a Pandas DataFrame.
2. Display the first and last 5 rows of the dataset.
3. Check the shape, data types, and missing values.
4. Generate summary statistics for numerical columns.
5. Find the average salary and total sales per department.
6. Identify the employee with the highest sales.
7. Detect outliers in the Salary column.
8. Plot a histogram for Age distribution.
9. Plot a bar chart for department-wise average salary.
10. Plot a scatter plot between Experience and Salary.
11. Check correlation between Salary and Sales.
12. Write 3 key insights from the analysis.